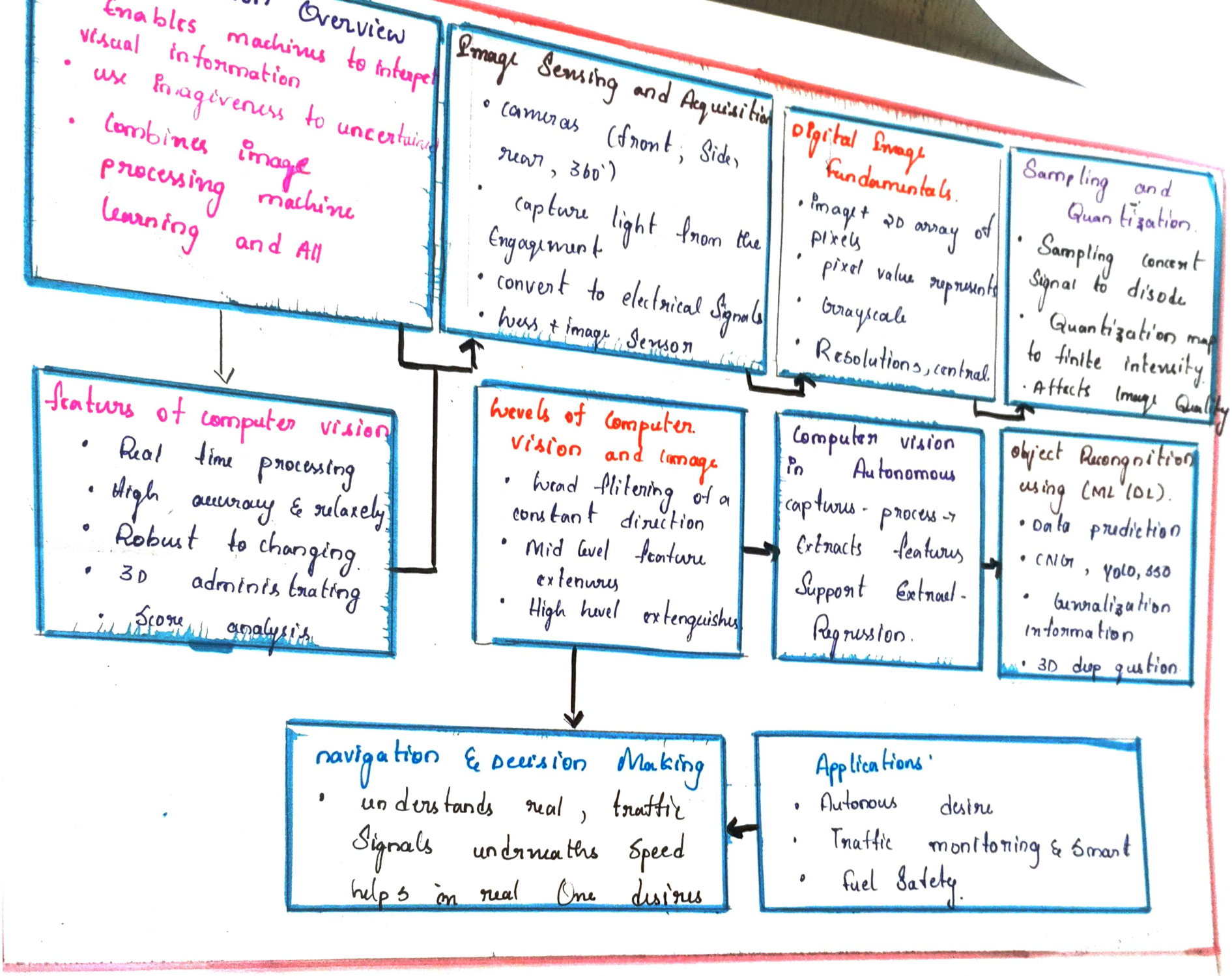
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course code :- ITA0502

course name :- Computer vision

[CO1 - Assignment Tool - 2].


Overview
Enables machines to interpret visual information
use imageveness to uncertain
Combines image processing machine learning and AI
Image Sensing and Acquisition
cameras (front, Side, rear, 360')
capture light from the Engagement
convert to electrical Signals
lens + image Sensor
Digital Image fundamentals
Image 2D array of pixels
pixel value represents
Grayscale
Resolutions, central
Sampling and Quantization
Sampling concert Signal to disode
Quantization map to finite intensity
Affects Image Quality
features of computer vision
Real time processing
High accuracy & relaxely
Robust to changing
3D administrating
Score analysis
levels of Computer vision and image
Head filtering of a constant direction
Mid Level feature extenuus
High level extenguishes
Computer vision in Autonomous
capturus - process → Extracts features
Support Extraal - Regression.
object Recognition using (ML (DL).
Data prediction
CNN, yolo, sso
Generalization information
3D dup gustion
navigation & decision Making
understands real, traffic Signals underneaths Speed
helps in real One desires
Applications:
Autonous desire
Traffic monitoring & Smart
fuel Safety.

computer vision Overview

- Enables machines intelligent
- helps detect lies

Support degross and treatment.

Image Sensing and Acquisitions.

- x-ray Sauces emits radiation
- detector files panel captures Signal.

Digital Image fundamentals

- Image = 2D array of pixels
- bumrative x-ray integral
- Refulction Contract.

Sampling and Quantization

- Sampling Continouous Signal. → pixels
- Higher in detech better and detail.

Levels of Computer vision processing

- how level mPssed enforcement
- High level contenton dassification Support.

Medical Imaging (x-ray)

circuit → cigaze → Process → default Academics.

Image processing & feature Extraction

- Edge detection. Segmentation
- feature Extractive, Shape pattern

Applications.

- detecting fracture infections
- Oral x-ray and Memography throat
- CT/MRI / hicensed

DIP (In Context)

- capturate Edge represention
- pixel intensity pissues
- Specilized frequency.

Image formation models

- x-ray interaction model like warnst (WIN)
- formates complexion murge

Overall Outcome

- clear enchancial dignified x-ray image
- Accurate and detection.